

Advanced Gene Mapping Course

January 11-15, 2027

The Rockefeller University

New York, NY

Emphasis: Analysis of sequence and other omics data. This course will cover both theory and applied exercises. Computer exercises will be carried out using a variety of programs (FaST-LMM, LDSC, Mediation, MR-PRESSO, PLINK, R, REGENIE, SuSiE, TwoSampleMR, etc).

Topics: Complex trait rare variant association studies, whole genome association studies (including sequence data), qualitative and quantitative traits, data quality control, functional annotation, detection of interactions, Transcriptome-wide association studies (TWAS), detection of pleiotropy, mediation analysis, Mendelian randomization, polygenic risk scores, fine mapping, heritability estimation, meta analysis, controlling for population admixture/substructure, detecting gene x gene interactions, and sample-size and power estimation.

Course Fee (academic): 200 USD

Travel stipends: For all course participants whose home institution is >20 miles from NYC

Instructors: Heather Cordell (University of Newcastle), Andrew DeWan (Yale University), Suzanne Leal (Rockefeller University & Columbia University), Shamil Sunyaev (Harvard University) & Gao Wang (Columbia University)

For additional information, course schedule and application form visit the course websites:

<https://statgen.us/Advgenemap2027>